



Irwansyah

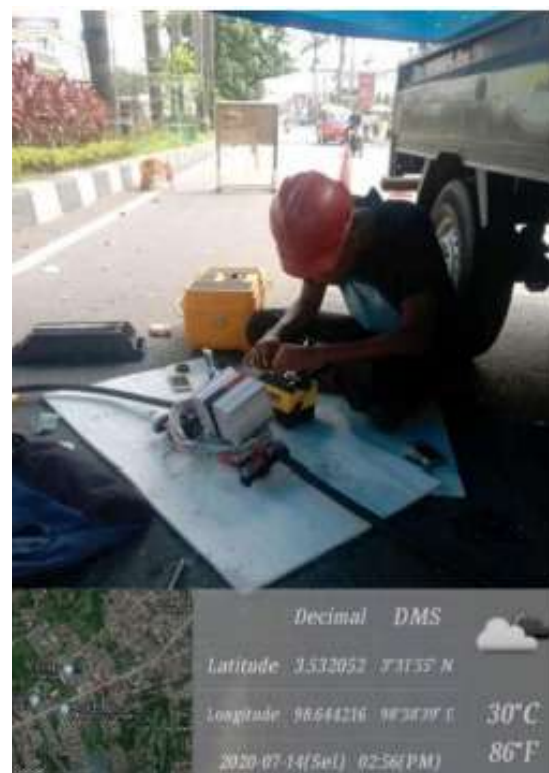
Telecommunication
Engineer



ABOUT ME



A Bachelor's degree graduate in Telecommunication Engineering from Telkom University with deep competence in network infrastructure and digital technology. Holds a professional certification as a Mikrotik Certified Network Associate (MTCNA) and is currently actively managing remote satellite infrastructure for the BAKTI AKSI project.



Possesses hands-on technical experience in the field, ranging from fiber optic termination, OSP mapping, to designing Deep Learning (CNN) models for biometrics. I combine field operational resilience with analytical data system intelligence.



Peralatan Data Saya

A robust infrastructure begins with precise planning and efficient technology.

Technical Skills)

- Telecommunication Network & Infrastructure
- Instrumentation & Calibration
- Programming & AI Implementation
- Data & Documentation

Tools

- VSAT Preventive & Corrective Maintenance (Hughes)
- Routing & Switching Protocols (RouterOS, Cisco IOS)
- Fiber Optic Splicing, OTDR, OPM Testing

JOBS



Executing the physical alignment, elevation adjustment, and fine-tuning of VSAT parabolic antennas to achieve the highest possible signal acquisition from the target satellite, ensuring stable internet access for remote public services.



Monitoring modem and network device statuses directly on-site to ensure optimal hardware health at public facilities



Executed critical Corrective Maintenance (CM) by diagnosing and replacing faulty VSAT transceivers (Hughes HB220-L), applying proper weatherproofing to connectors, and re-calibrating the antenna to restore optimal transmission (TX) and reception (RX) signals.

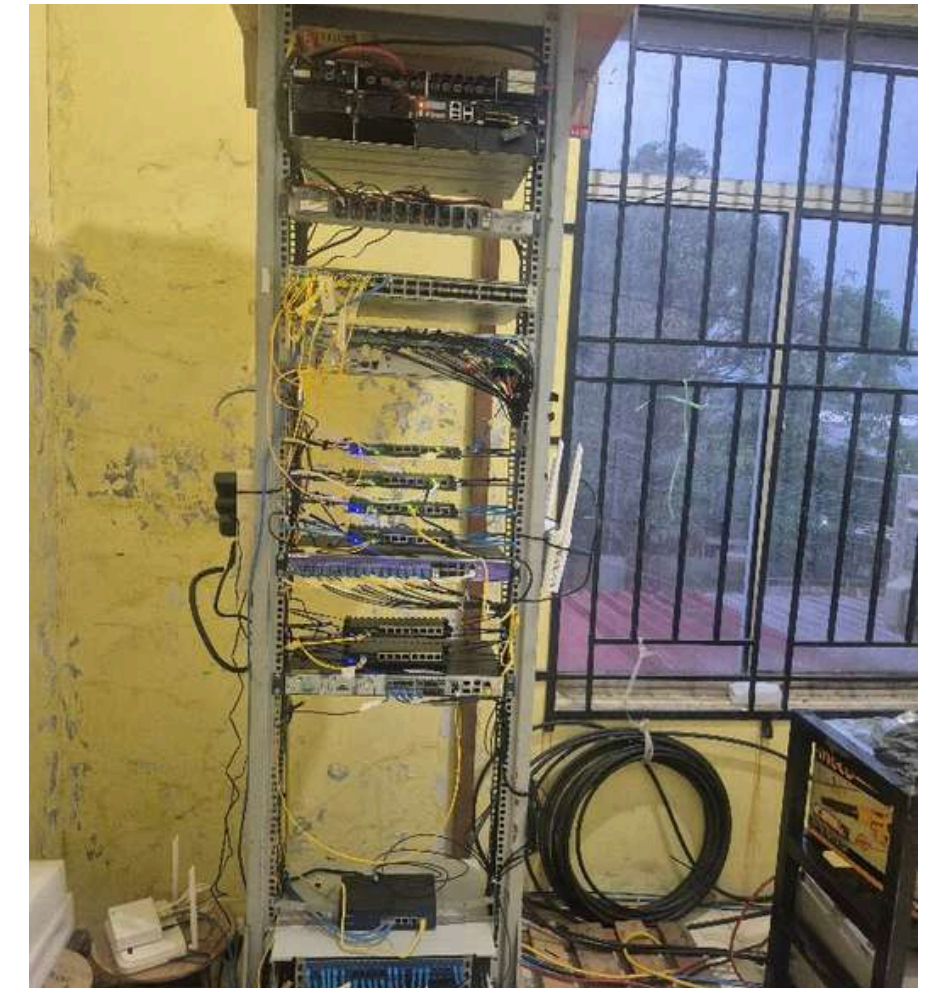
AS A VSAT ENGINEER ON SITE



Executed aerial cable pulling, pole mounting, and home cable installation (IKR), adhering to safety protocols while establishing robust physical network infrastructure for new and existing clients.

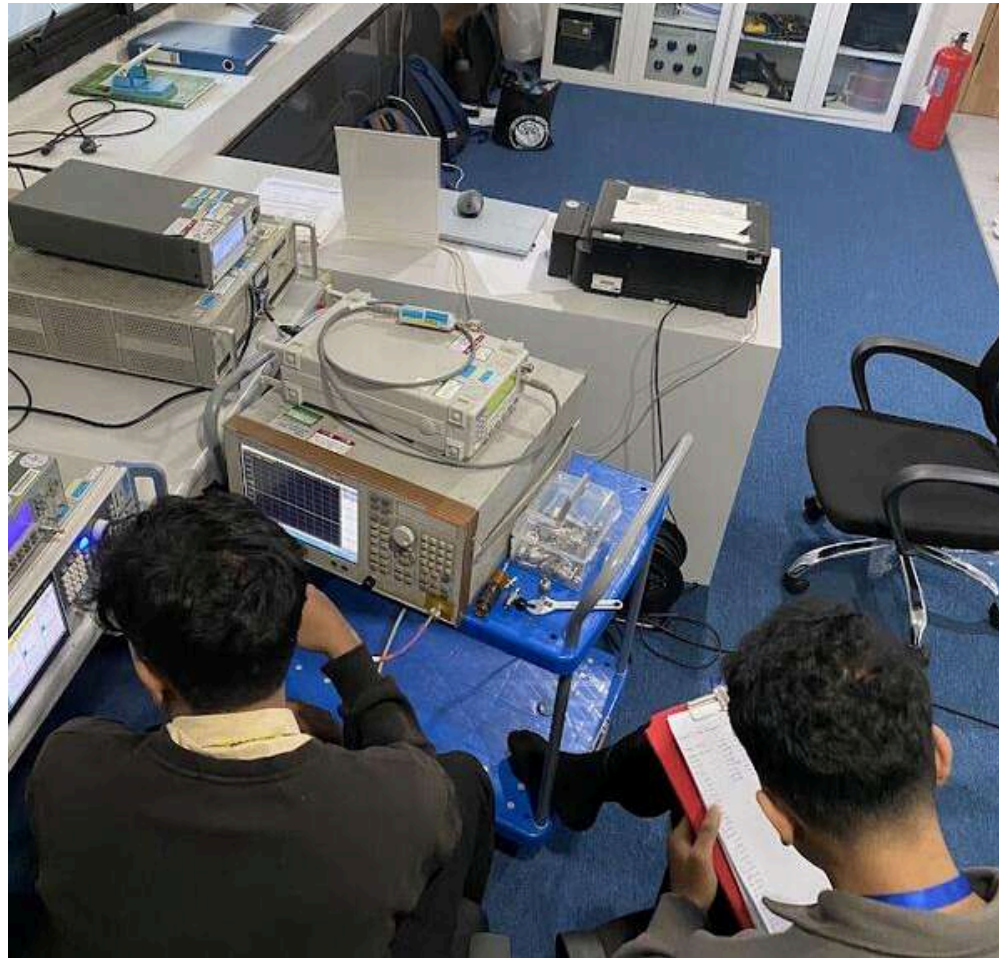


Performed precision fusion splicing and core preparation using specialized equipment (fiber splicer and cleaver) for routine maintenance.

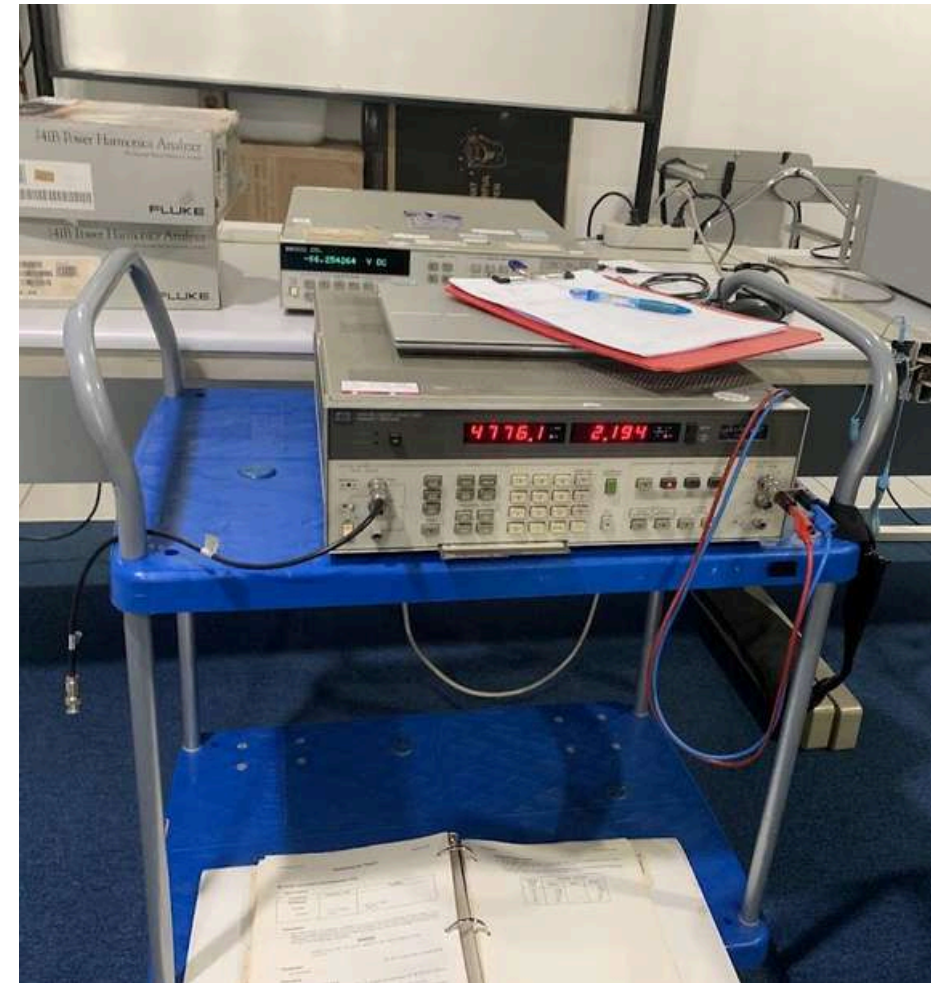


Assisted in installing, patching, and organizing fiber and Ethernet cabling within main distribution server racks while collaborating closely with field teams and NOC personnel to resolve physical and logical network outages through rapid-response troubleshooting.

AS A TECHNICAL SUPPORT



Processed raw measurement data into structured Excel reports to ensure zero-error reporting, enabling smooth administrative workflows for official Telkom Indonesia calibration certificate issuance.



Operated high-precision reference standards—including HP 3458A 8½-Digit Multimeters, Audio Analyzers, and Power Harmonics Analyzers—to measure and verify voltage, current, and signal accuracy.



Executed functional checks, maintenance, and periodic calibration on advanced fiber optic testing platforms (such as EXFO FTBx / OTDR units), updating official calibration labels upon passing quality compliance.

AS A TEST ENGINEER



Executed high-capacity fiber optic duct cable splicing and core management for 288-core fiber infrastructure in the field, utilizing precision fusion splicer equipment to ensure low insertion loss, optimal signal integrity, and high-bandwidth transmission stability across the entire core capacity.



Executed cross-connect operations, maintenance, and patch cord organization within outdoor ODC cabinets—managing passive splitters and distribution trays to ensure stable signal allocation toward end-user lines.



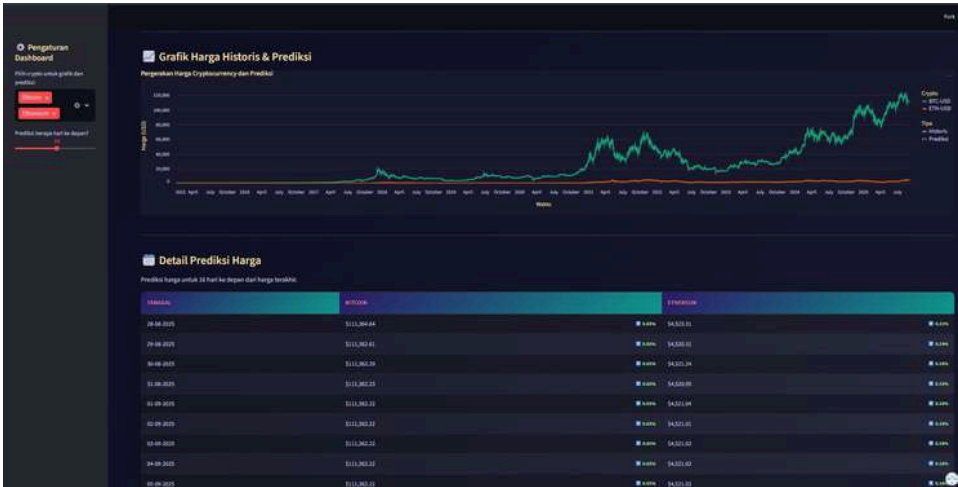
Managed and executed Outside Plant (OSP) infrastructure deployments, including aerial fiber optic cable pulling, pole installation, ascending working platforms safely, and proper cable securing adhering to safety protocols.

AS A TECHNICAL SUPPORT

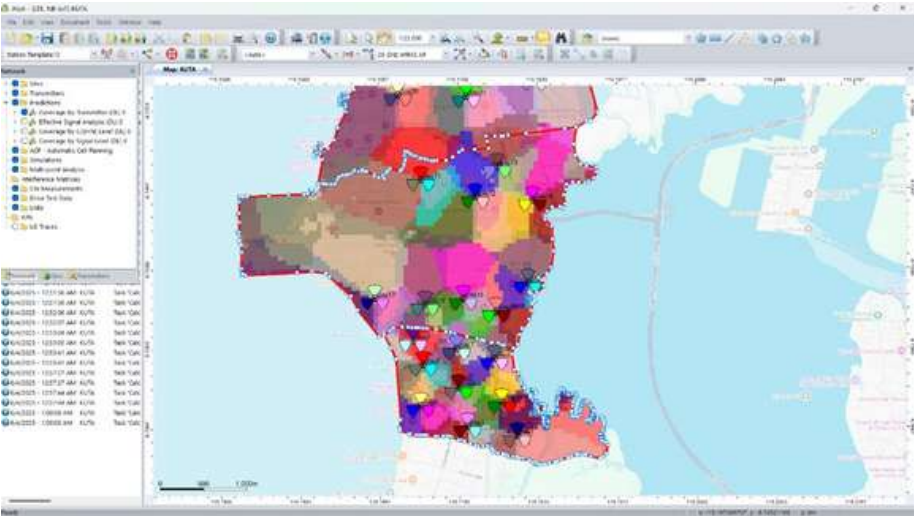
Featured Works



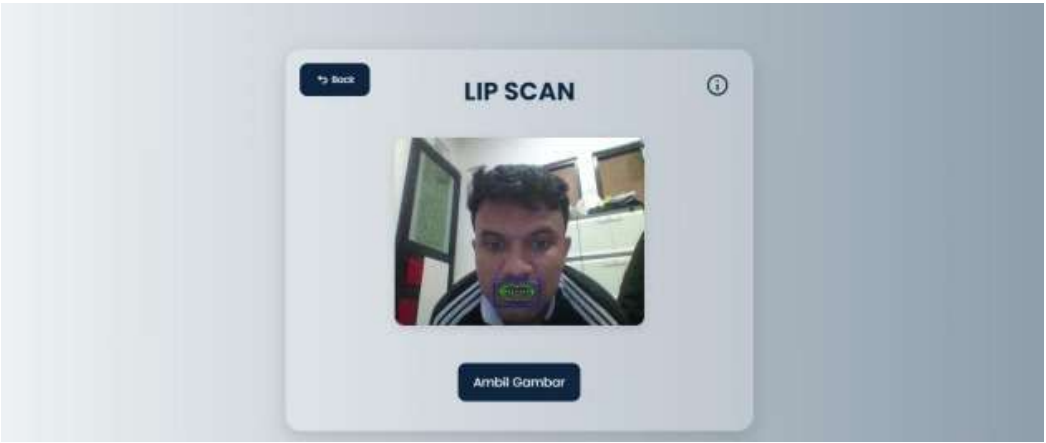
Cryptocurrency Price Analysis and Prediction Dashboard



Atoll Cell Coverage Design



Individual Detection Based on Digital Image Processing



DESCRIPTION



Atoll Cell Coverage Design

Conducted simulation and coverage area planning for the Kuta Subdistrict, Badung, using Atoll, focusing on physical antenna parameter optimization for optimal signal quality standards.



Cryptocurrency Price Analysis and Prediction Dashboard

Developed a Streamlit Cloud and ARIMA-based application for 30-day price prediction, equipped with interactive visualizations and real-time data tracking.



Individual Detection Based on Digital Image Processing

Developed a Deep Learning model based on the MobileNetV2 architecture that achieved perfect accuracy (100%) in identifying 20 different individuals from a total of 200 image datasets.

CASE STUDY



Challenge

Designing optimal 4G LTE signal coverage that meets operator KPI standards in a densely populated geographical area (Kuta, Bali).

My Role:
RF Planner

Process

- Using Atoll software to map signal propagation based on digital geographical data.
- Determining the best site locations and configuring antenna parameters (such as Azimuth and Mechanical/Electrical Tilt).
- Running prediction simulations to evaluate signal strength and interference.

Result

Produced a network design that meets industry KPI standards (RSRP > -100 dBm and SINR > 15 dB). This design ensures stable connectivity and even coverage to support high data traffic in the tourist area.

Network Planning



Challenge

Training an AI model to identify 20 different individuals with high accuracy, despite using a very limited dataset (200 images).

Peran Saya:
Deep Learning Engineer

Process

- Using Python (NumPy & Pandas) for digital image normalization and data augmentation to enrich dataset variation.
- Implementing the MobileNetV2 architecture (Transfer Learning) which is efficient for computation, reconfigured for specific 20-class classification.
- Performing hyperparameter tuning and cross-validation to ensure the model recognizes unique lip print patterns.

Result

Successfully achieved a 100% accuracy rate on test data. The system is able to accurately distinguish 20 individuals, proving the effectiveness of the Convolutional Neural Network (CNN) method for modern biometric security applications.

Deep Learning & Image Processing